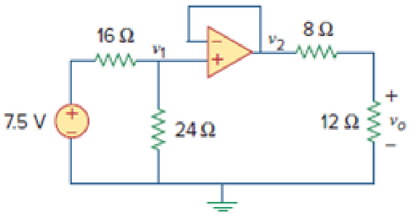
Problem

Find *vo* in the op amp circuit of Fig. 5.65.

Figure. 5.65:



Step-by-step solution

Step 1 of 3

Draw the circuit diagram as shown in Figure 1.

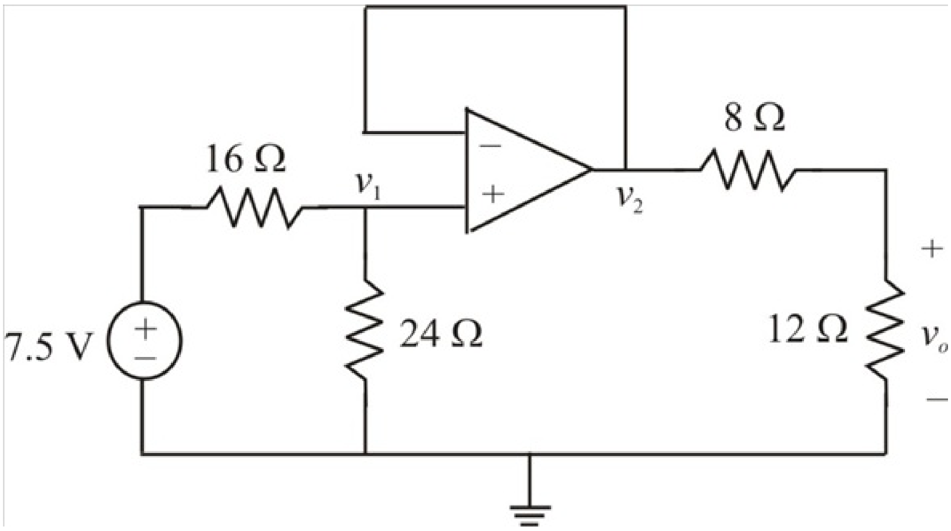


Figure 1

Step 2 of 3

The circuit is labeled as shown in Figure 2.

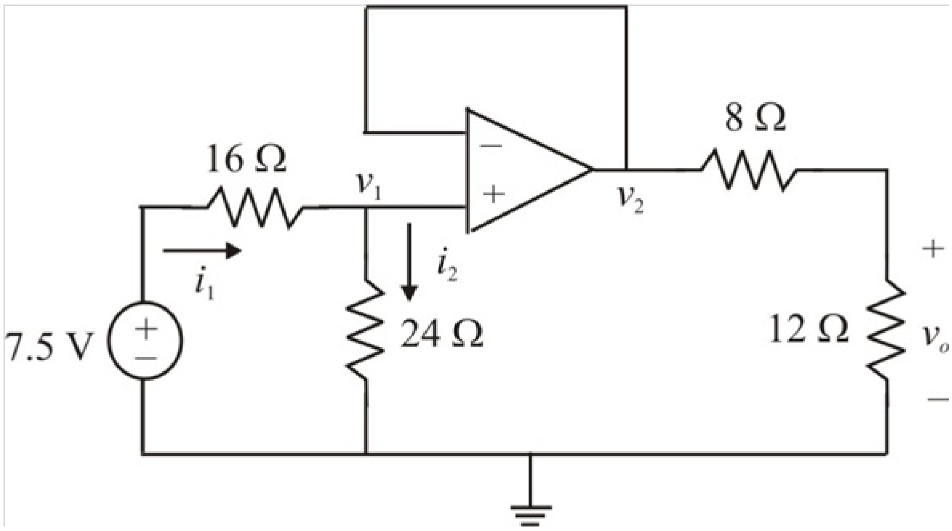


Figure 2

Step 3 of 3

Applying Kirchhoff's current law to node 1,

$$
\begin{aligned}
i_1 &= i_2 \\
\frac{7.5 - v_1}{16} &= \frac{v_1 - 0}{24} \\
22.5 - 3v_1 &= 2v_1 \\
5v_1 &= 22.5 \\
v_1 &= 4.5\text{ V}
\end{aligned}
$$

But, $v_1 = v_2$ for an ideal op amp

Therefore voltage $v_2 = 4.5\text{ V}$

Applying voltage division,

$$
\begin{aligned}
v_o &= \frac{12}{12+8}(v_2) \\
&= \frac{12}{12+8}(4.5) \\
&= 2.7\text{ V}
\end{aligned}
$$

Therefore the voltage v_o is $\boxed{2.7\text{ V}}$.